Qualification Pack

Mechatronics

QP Code: ELE/Q7108

Version: 2.0

NSQF Level: 4.5

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Contents

ELE/Q7108: Mechatronics

Brief Job Description

The Mechatronics qualification develops integrated skills in mechanics, electronics, pneumatics, hydraulics, and computer technologies. It prepares technicians to design, build, program, and maintain automated industrial systems, operate HMIs, manage fluid power, and work safely with advanced technologies like RFID, NFC, and vision systems.

Personal Attributes

The individual must have managerial and problem-solving skills along with the ability to take independent decisions. The person must also have excellent interpersonal and organisational skills along with good written and verbal communication skills. The individual must be able to coordinate with multiple parties to achieve the work objectives.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. ELE/N7124: Work organization and management (Mechatronics)
2. ELE/N7125: Communication and interpersonal skills (Mechatronics)
3. ELE/N7126: Developing mechatronics systems (Mechatronics)
4. ELE/N7127: Using industrial controllers (Mechatronics)
5. ELE/N7128: Software programming (Mechatronics)
6. ELE/N7129: Circuit schematic (Mechatronics)
7. ELE/N7130: Analysing, commissioning, and maintenance (Mechatronics)

Qualification Pack (QP) Parameters

Sector	Electronics
Sub-Sector	Industrial Automation
Occupation	Engineering-I&A
Country	India

Qualification Pack

NSQF Level	4.5
Credits	17
Aligned to NCO/ISCO/ISIC Code	NCO-2015/7412.0101
Minimum Educational Qualification & Experience	No formal education prescribed
Minimum Level of Education for Training in School	Not Applicable
Pre-Requisite License or Training	NA
Minimum Job Entry Age	14 Years
Last Reviewed On	NA
Next Review Date	06/02/2028
NSQC Approval Date	06/02/2026
Version	2.0
Reference code on NQR	QG-4.5-EH-05198-2026-V1-ESSCI
NQR Version	2

ELE/N7124: Work organization and management (Mechatronics)

Description

This NOS unit is about proper training, the correct use of equipment, and adherence to safety protocols in manufacturing settings.

Scope

The scope covers the following :

- Principles and Applications of Safe Working
- Purposes, Uses, Care, and Maintenance of Equipment and Materials
- Environmental and Safety Principles in Good Housekeeping
- Principles of Team Working and Their Applications

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

PC1. Maintain a safe, clean, and well-organized work area following workplace design principles.
PC2. Prepare self for tasks by adhering to all relevant health, safety, and environmental guidelines.
PC3. Schedule work activities efficiently to maximize productivity and minimize disruptions.
PC4. Select, handle, and operate equipment and materials safely as per manufacturer instructions.
PC5. Apply workplace health, safety, and environmental standards consistently during all activities.
PC6. Restore the work area to an acceptable condition after completing tasks.
PC7. Collaborate effectively with team members to achieve group objectives.
PC8. Exchange constructive feedback and offer support to colleagues to enhance team performance.
PC9. Work efficiently to minimize wastage of materials, time, and energy.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. Knowledge of workplace organization and housekeeping principles.
KU2. Understanding of health, safety, and environmental guidelines.
KU3. Knowledge of work scheduling and productivity optimization methods.
KU4. Knowledge of work scheduling and productivity optimization methods.
KU5. Understanding of safe handling and operation of equipment and materials.

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. Ability to maintain a clean, safe, and organized work environment.

GS2. Skill in preparing for tasks while following safety and environmental guidelines.
GS3. Ability to plan and schedule work efficiently.
GS4. Skill in safely handling and operating tools and materials.
GS5. Ability to consistently apply workplace safety standards.

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	-	**10**	**-**	**-**
PC1. Maintain a safe, clean, and well-organized work area following workplace design principles.	-	2	-	-
PC2. Prepare self for tasks by adhering to all relevant health, safety, and environmental guidelines.	-	1	-	-
PC3. Schedule work activities efficiently to maximize productivity and minimize disruptions.	-	1	-	-
PC4. Select, handle, and operate equipment and materials safely as per manufacturer instructions.	-	1	-	-
PC5. Apply workplace health, safety, and environmental standards consistently during all activities.	-	1	-	-
PC6. Restore the work area to an acceptable condition after completing tasks.	-	1	-	-
PC7. Collaborate effectively with team members to achieve group objectives.	-	1	-	-
PC8. Exchange constructive feedback and offer support to colleagues to enhance team performance.	-	1	-	-
PC9. Work efficiently to minimize wastage of materials, time, and energy.	-	1	-	-
NOS Total	**-**	**10**	**-**	**-**

National Occupational Standards (NOS) Parameters

NOS Code	ELE/N7124
NOS Name	Work organization and management (Mechatronics)
Sector	Electronics
Sub-Sector	
Occupation	Engineering-I&A
NSQF Level	4.5
Credits	2
Version	2.0
Last Reviewed Date	06/02/2026
Next Review Date	06/02/2028
NSQC Clearance Date	06/02/2026

ELE/N7125: Communication and interpersonal skills (Mechatronics)

Description

This NOS unit is about the communication and interpersonal skills

Scope

The scope covers the following :

- Range and Purposes of Documentation and Publications in Electronic Forms
- Technical Language Associated with the Skill and Technology
- Standards for Routine and Exception Reporting in Oral and Electronic Form
- Required Standards for Communicating with Clients, Team Members, and Others

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

PC1. Read, interpret, and extract technical information from documentation in paper or electronic formats.

PC2. Communicate clearly and effectively using oral, written, and electronic methods.

PC3. Use standard communication technologies and tools for professional interactions.

PC4. Explain and discuss complex technical concepts, principles, and applications with colleagues and stakeholders.

PC5. Prepare and complete routine and exception reports accurately as per organizational standards.

PC6. Respond to client needs promptly and professionally, both in person and through indirect communication.

PC7. Collect relevant information and prepare required documentation according to client or organizational requirements.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. Knowledge of interpreting technical documentation in both paper and electronic formats.

KU2. Understanding of effective communication methods in oral, written, and digital forms.

KU3. Knowledge of communication tools and technologies used in professional environments.

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. Ability to read, interpret, and extract relevant technical information accurately.

GS2. Skill in communicating clearly and effectively across different mediums.

GS3. Skill in communicating clearly and effectively across different mediums.

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	-	**10**	**-**	**-**
PC1. Read, interpret, and extract technical information from documentation in paper or electronic formats.	-	2	-	-
PC2. Communicate clearly and effectively using oral, written, and electronic methods.	-	2	-	-
PC3. Use standard communication technologies and tools for professional interactions.	-	2	-	-
PC4. Explain and discuss complex technical concepts, principles, and applications with colleagues and stakeholders.	-	1	-	-
PC5. Prepare and complete routine and exception reports accurately as per organizational standards.	-	1	-	-
PC6. Respond to client needs promptly and professionally, both in person and through indirect communication.	-	1	-	-
PC7. Collect relevant information and prepare required documentation according to client or organizational requirements.	-	1	-	-
NOS Total	**-**	**10**	**-**	**-**

National Occupational Standards (NOS) Parameters

NOS Code	ELE/N7125
NOS Name	Communication and interpersonal skills (Mechatronics)
Sector	Electronics
Sub-Sector	
Occupation	Engineering-I&A
NSQF Level	4.5
Credits	2
Version	2.0
Last Reviewed Date	06/02/2026
Next Review Date	06/02/2028
NSQC Clearance Date	06/02/2026

ELE/N7126: Developing mechatronics systems (Mechatronics)

Description

This NOS unit is about design and optimize systems for industrial applications, resolving uncertainties and ensuring compliance with specifications.

Scope

The scope covers the following :

- Carry Out Systems Design for Given Industrial Applications
- Identify and Resolve Areas of Uncertainty within the Briefs or Specifications
- Optimize the Design within the Parameters of the Specification
- Assemble Machines According to Documentation

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

PC1. Carry out system design for given industrial applications, considering functional and technical requirements.

PC2. Identify and resolve areas of uncertainty within briefs or specifications.

PC3. Optimize designs within the parameters of the specifications and operational constraints.

PC4. Assemble machines according to digital documentation, including written instructions and 2D/3D designs.

PC5. Connect wires, tubing, and fluid lines according to industry standards.

PC6. Incorporate robots and handling systems within mechatronic systems as required.

PC7. Integrate HMI devices and interfaces into the system for effective operation.

PC8. Incorporate safety devices such as emergency stops, safety sensors, and relays into the system.

PC9. Install, set up, and adjust mechanical, pneumatic, electrical, and sensor systems to ensure full functionality.

PC10. Configure and parameterize complex sensors, including vision systems, color sensors, and incremental systems, following standard manuals.

PC11. Commission machines using auxiliary equipment and PLCs, adhering to standard procedures and documentation.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. Knowledge of system design principles for industrial and mechatronic applications.

KU2. Understanding of identifying and resolving ambiguities in technical specifications.

KU3. Knowledge of design optimization within operational and technical constraints.

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. Ability to design systems based on industrial application requirements.
GS2. Skill in identifying and resolving specification gaps effectively.
GS3. Ability to optimize designs within given constraints.

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	-	**15**	**-**	**-**
PC1. Carry out system design for given industrial applications, considering functional and technical requirements.	-	2	-	-
PC2. Identify and resolve areas of uncertainty within briefs or specifications.	-	2	-	-
PC3. Optimize designs within the parameters of the specifications and operational constraints.	-	2	-	-
PC4. Assemble machines according to digital documentation, including written instructions and 2D/3D designs.	-	2	-	-
PC5. Connect wires, tubing, and fluid lines according to industry standards.	-	1	-	-
PC6. Incorporate robots and handling systems within mechatronic systems as required.	-	1	-	-
PC7. Integrate HMI devices and interfaces into the system for effective operation.	-	1	-	-
PC8. Incorporate safety devices such as emergency stops, safety sensors, and relays into the system.	-	1	-	-
PC9. Install, set up, and adjust mechanical, pneumatic, electrical, and sensor systems to ensure full functionality.	-	1	-	-
PC10. Configure and parameterize complex sensors, including vision systems, color sensors, and incremental systems, following standard manuals.	-	1	-	-
PC11. Commission machines using auxiliary equipment and PLCs, adhering to standard procedures and documentation.	-	1	-	-
NOS Total	**-**	**15**	**-**	**-**

National Occupational Standards (NOS) Parameters

NOS Code	ELE/N7126
NOS Name	Developing mechatronics systems (Mechatronics)
Sector	Electronics
Sub-Sector	
Occupation	Engineering-I&A
NSQF Level	4.5
Credits	2
Version	2.0
Last Reviewed Date	06/02/2026
Next Review Date	06/02/2028
NSQC Clearance Date	06/02/2026

ELE/N7127: Using industrial controllers (Mechatronics)

Description

This NOS unit is about integrate PLCs into mechatronics systems by configuring industrial controllers, control circuitry, and communication protocols

Scope

The scope covers the following :

- Integrate and Connect PLCs to Mechatronics Systems
- Set-up an Industrial Network/Bus System for Communication
- Make the Necessary Configurations of Industrial Controllers
- Configure All Aspects of PLCs as Required

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

PC1. Integrate and connect PLCs to mechatronic systems for control and automation.

PC2. Set up industrial networks/bus systems to enable communication between PLCs, HMI devices, other electronics, distributed devices, and PCs via OPC-UA.

PC3. Configure industrial controllers to meet operational requirements.

PC4. Set up and configure all aspects of PLCs, including associated control circuitry, for correct system operation.

PC5. Establish and verify proper communications between and among controllers and peripheral devices.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. Knowledge of PLC integration and its role in mechatronic system automation.

KU2. Understanding of industrial networks and communication protocols such as OPC-UA.

KU3. Knowledge of configuration and operation of industrial controllers.

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. Ability to integrate and connect PLCs with mechatronic systems.

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	-	**20**	**-**	**-**
PC1. Integrate and connect PLCs to mechatronic systems for control and automation.	-	4	-	-
PC2. Set up industrial networks/bus systems to enable communication between PLCs, HMI devices, other electronics, distributed devices, and PCs via OPC-UA.	-	4	-	-
PC3. Configure industrial controllers to meet operational requirements.	-	4	-	-
PC4. Set up and configure all aspects of PLCs, including associated control circuitry, for correct system operation.	-	4	-	-
PC5. Establish and verify proper communications between and among controllers and peripheral devices.	-	4	-	-
NOS Total	**-**	**20**	**-**	**-**

National Occupational Standards (NOS) Parameters

NOS Code	ELE/N7127
NOS Name	Using industrial controllers (Mechatronics)
Sector	Electronics
Sub-Sector	
Occupation	Engineering-I&A
NSQF Level	4.5
Credits	4
Version	2.0
Last Reviewed Date	06/02/2026
Next Review Date	06/02/2028
NSQC Clearance Date	06/02/2026

ELE/N7128: Software programming (Mechatronics)

Description

This NOS unit is about control mechatronic systems, including PLCs, HMI devices, and signal processing.

Scope

The scope covers the following :

- Write Programs to Control a Mechatronic System
- Visualize the Process and Operation Using Software
- Programme PLCs, Including Digital and Analogue Signal Processing and Industrial Field Buses
- Programme HMI Devices

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

PC1. Write software programs to control mechatronic systems, ensuring correct functionality.

PC2. Visualize and simulate processes and operations using industrial software tools.

PC3. Program PLCs, including handling digital and analog signals and integrating industrial field buses.

PC4. Develop and configure HMI devices to provide interactive control and monitoring.

PC5. Implement reliable and correct communication (“handshakes”) between PLCs and other system components.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1.
- Knowledge of PLC integration and its role in mechatronic system automation.
- Understanding of industrial networks and communication protocols such as OPC-UA.
- Knowledge of configuration and operation of industrial controllers.
- Understanding of PLC setup, control circuitry, and system integration requirements.
- Knowledge of communication systems between controllers and peripheral devices.

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1.
- Ability to integrate and connect PLCs with mechatronic systems.
- Skill in setting up industrial networks for system communication.
- Ability to configure controllers as per operational requirements.
- Skill in setting up and configuring PLC systems and control circuits.
- Ability to establish and verify communication between controllers and connected devices.

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	-	**20**	**-**	**-**
PC1. Write software programs to control mechatronic systems, ensuring correct functionality.	-	4	-	-
PC2. Visualize and simulate processes and operations using industrial software tools.	-	4	-	-
PC3. Program PLCs, including handling digital and analog signals and integrating industrial field buses.	-	4	-	-
PC4. Develop and configure HMI devices to provide interactive control and monitoring.	-	4	-	-
PC5. Implement reliable and correct communication ("handshakes") between PLCs and other system components.	-	4	-	-
NOS Total	**-**	**20**	**-**	**-**

National Occupational Standards (NOS) Parameters

NOS Code	ELE/N7128
NOS Name	Software programming (Mechatronics)
Sector	Electronics
Sub-Sector	
Occupation	Engineering-I&A
NSQF Level	4.5
Credits	4
Version	2.0
Last Reviewed Date	06/02/2026
Next Review Date	06/02/2028
NSQC Clearance Date	06/02/2026

ELE/N7129: Circuit schematic (Mechatronics)

Description

This NOS unit is about reading and interpreting pneumatic, hydraulic, and electrical circuit schematics for system design.

Scope

The scope covers the following :

- Read and Use Pneumatic, Hydraulic, and Electrical Circuit Schematics
- Design the Circuits Using Modern Software Tools

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

PC1. Understand and apply principles, applications, and standards for circuit schematics.

PC2. Read and interpret pneumatic, hydraulic, and electrical circuit schematics accurately.

PC3. Design electrical circuits for mechatronic systems using standard principles and practices.

PC4. Assemble and implement designed circuits in mechatronic systems following industry standards.

PC5. Use modern software tools to design, simulate, and validate circuit schematics.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1.
- Knowledge of principles, applications, and standards for circuit schematics.
- Understanding of pneumatic, hydraulic, and electrical circuit diagrams.
- Knowledge of electrical circuit design for mechatronic systems.
- Understanding of circuit assembly and implementation as per industry standards.
- Knowledge of modern software tools for circuit design, simulation, and validation.

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1.
- Ability to apply schematic principles and standards in circuit design.
- Skill in reading and interpreting circuit schematics accurately.
- Ability to design electrical circuits for mechatronic applications.
- Skill in assembling and implementing circuits as per standards.
- Ability to use software tools for designing, simulating, and validating circuits.

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	-	**10**	**-**	**-**
PC1. Understand and apply principles, applications, and standards for circuit schematics.	-	2	-	-
PC2. Read and interpret pneumatic, hydraulic, and electrical circuit schematics accurately.	-	2	-	-
PC3. Design electrical circuits for mechatronic systems using standard principles and practices.	-	2	-	-
PC4. Assemble and implement designed circuits in mechatronic systems following industry standards.	-	2	-	-
PC5. Use modern software tools to design, simulate, and validate circuit schematics.	-	2	-	-
NOS Total	**-**	**10**	**-**	**-**

National Occupational Standards (NOS) Parameters

NOS Code	ELE/N7129
NOS Name	Circuit schematic (Mechatronics)
Sector	Electronics
Sub-Sector	
Occupation	Engineering-I&A
NSQF Level	4.5
Credits	1
Version	2.0
Last Reviewed Date	06/02/2026
Next Review Date	06/02/2028
NSQC Clearance Date	06/02/2026

ELE/N7130: Analysing, commissioning, and maintenance (Mechatronics)

Description

This NOS unit is about testing and analyze individual modules and complete mechatronic systems against criteria, identifying faults and optimizing performance.

Scope

The scope covers the following :

- Test Run Individual Modules and Assembled Systems
- Review Each Part of the Process Against Established Criteria
- Find Faults in a Mechatronic System Using Appropriate Analytical Techniques
- Collect Data with Representations as Diagrams (Digitalization)

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

PC1. Conduct test runs of individual modules and fully assembled mechatronic systems.
PC2. Review each part of the process and system against established performance criteria.
PC3. Identify faults in mechatronic systems using appropriate analytical and problem-solving techniques.
PC4. Collect and represent operational data using diagrams, bar graphs, and digital formats.
PC5. Perform repairs on components efficiently and effectively.
PC6. Optimize the operation of individual modules using analysis and problem-solving methods.
PC7. Optimize the operation of the entire mechatronic system for performance and efficiency.
PC8. Apply principles and techniques to generate creative and innovative solutions for system issues.
PC9. Present assembled systems to clients and respond to queries professionally.
PC10. Measure and monitor consumed energy using sensors and appropriate measuring devices.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1.
- Knowledge of testing procedures for modules and complete mechatronic systems.
- Understanding of performance evaluation criteria and system review methods.
- Knowledge of fault identification and analytical troubleshooting techniques.
- Understanding of data collection and representation methods using graphs and digital tools.
- Knowledge of repair techniques for mechatronic components.
- Understanding of module-level optimization methods.
- Knowledge of system-level optimization for performance and efficie

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1.
- Ability to conduct test runs of modules and complete systems effectively.
- Skill in reviewing system performance against defined criteria.
- Ability to identify and diagnose faults using analytical methods.
- Skill in collecting and presenting operational data clearly.
- Ability to perform repairs efficiently on system components.
- Skill in optimizing individual modules for improved performance.
- Ability to optimize overall system efficiency and functionality.
- Skill in applying creative thinking to solve

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	-	**15**	**-**	**-**
PC1. Conduct test runs of individual modules and fully assembled mechatronic systems.	-	2	-	-
PC2. Review each part of the process and system against established performance criteria.	-	2	-	-
PC3. Identify faults in mechatronic systems using appropriate analytical and problem-solving techniques.	-	2	-	-
PC4. Collect and represent operational data using diagrams, bar graphs, and digital formats.	-	2	-	-
PC5. Perform repairs on components efficiently and effectively.	-	2	-	-
PC6. Optimize the operation of individual modules using analysis and problem-solving methods.	-	1	-	-
PC7. Optimize the operation of the entire mechatronic system for performance and efficiency.	-	1	-	-
PC8. Apply principles and techniques to generate creative and innovative solutions for system issues.	-	1	-	-
PC9. Present assembled systems to clients and respond to queries professionally.	-	1	-	-
PC10. Measure and monitor consumed energy using sensors and appropriate measuring devices.	-	1	-	-
NOS Total	**-**	**15**	**-**	**-**

National Occupational Standards (NOS) Parameters

NOS Code	ELE/N7130
NOS Name	Analysing, commissioning, and maintenance (Mechatronics)
Sector	Electronics
Sub-Sector	
Occupation	Engineering-I&A
NSQF Level	4.5
Credits	2
Version	2.0
Last Reviewed Date	06/02/2026
Next Review Date	06/02/2028
NSQC Clearance Date	06/02/2026

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Criteria for assessment for each Qualification Pack will be created by the Sector Skill Council. Each Performance Criteria (PC) (PC) will be assigned marks proportional to its importance in NOS. SSC will also lay down proportion of marks for Theory and Skills Practical for each PC.

2. The assessment for the theory part will be based on knowledge bank of questions created by the SSC.

3. Individual assessment agencies will create unique question papers for theory part for each candidate at each examination/training centre (as per assessment criteria below).

4. Individual assessment agencies will create unique evaluations for skill practical for every student at each examination/ training centre based on these criteria.

5. In case of successfully passing only certain number of NOSs, the trainee is eligible to take subsequent assessment on the balance NOS's to pass the Qualification Pack.

6. In case of unsuccessful completion, the trainee may seek reassessment on the Qualification Pack

Minimum Aggregate Passing % at QP Level : 70

(**Please note**: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
ELE/N7124.Work organization and management (Mechatronics)	-	10	-	-	10	10
ELE/N7125.Communication and interpersonal skills (Mechatronics)	-	10	-	-	10	10
ELE/N7126.Developing mechatronics systems (Mechatronics)	-	15	-	-	15	15
ELE/N7127.Using industrial controllers (Mechatronics)	-	20	-	-	20	20
ELE/N7128.Software programming (Mechatronics)	-	20	-	-	20	20
ELE/N7129.Circuit schematic (Mechatronics)	-	10	-	-	10	10
ELE/N7130.Analysing, commissioning, and maintenance (Mechatronics)	-	15	-	-	15	15
Total	**0**	**100**	**-**	**-**	**100**	**100**

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.